

1. Administrative & Study Identification

Full Study Title: A Single Arm, Open Label Study to Assess the Efficacy, Safety and Tolerability of Once-monthly Administration of Subcutaneous C.E.R.A. for the Treatment of Chronic Renal Anaemia in Pre-dialysis Patients Not Currently Treated With ESA. **Short Title/Acronym:** A Study of Monthly Subcutaneous Mircera for the Treatment of Chronic Renal Anemia in Predialysis Patients Not Treated With ESA. **Protocol Number:** ML21524 **NCT Number:** NCT00661388 **Posting ID:** 505575565193260733 **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Methoxy polyethylene glycol-epoetin beta (Mircera) / Continuous erythropoietin receptor activator (C.E.R.A.) **ATC Code:** B03XA03 **Phase of Development:** Phase 3 **Trial Status:** COMPLETED (Study has been completed) **Medical Monitor:** Hoffmann-La Roche Clinical Trial Manager

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of subcutaneous methoxy polyethylene glycol-epoetin beta (Mircera) for the correction and maintenance of hemoglobin levels in predialysis patients with renal anemia who are not currently treated with erythropoietin stimulating agents (ESA). **Secondary Objectives:** To evaluate the overall tolerability and necessary dose adjustments of once-monthly administration of subcutaneous C.E.R.A. to maintain target hemoglobin levels.

2.2 Background & Rationale To address the clinical need for effective and convenient anemia management in chronic kidney disease (CKD) patients in the predialysis stage who are not currently treated with erythropoietin-stimulating agents (ESAs). The use of a once-monthly subcutaneous injection aims to simplify the treatment burden while steadily maintaining hemoglobin within the optimal target range (10.0 - 12.0 g/dL).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Status:** COMPLETED

3.2 Planned Enrollment & Duration Target N: 75 participants (Note: Summary text indicates an anticipated target sample size of 100-500 individuals) **Number of Sites:** Multiple Sites **Estimated**

Study Start: August 01, 2008 **Estimated Primary Completion:** December 31, 2010 (Study is now COMPLETED)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult patients, ≥ 18 years of age. 2. Confirmed diagnosis of chronic renal anemia in the predialysis stage. 3. No ESA (erythropoietin-stimulating agent) therapy during the previous 3 months.

4.2 Exclusion Criteria 1. Transfusion of red blood cells during the previous 2 months. 2. Poorly controlled hypertension requiring hospitalization in the previous 6 months. 3. Significant acute or chronic bleeding.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Product Reference: [EMA Product Information](#) **Dosage/Frequency:** Initial dose of C.E.R.A. at $1.2\text{ }\mu\text{g/kg}$ administered once every 4 weeks. Subsequent doses will be adjusted at the clinician's discretion to maintain the participant's hemoglobin within the target range of 10.0 g/dL and 12.0 g/dL . **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 44 weeks (with an anticipated time on study treatment of 3 to 12 months).

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for chronic kidney disease and stable hypertension. **Prohibited:** Transfusion of red blood cells (unless medically necessary for rescue, which may affect endpoint evaluation) and other non-study ESAs.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -14 to 0	Informed Consent, Medical History, Baseline Hemoglobin & Blood Pressure Labs
Baseline	Day 0	First Dose of C.E.R.A. ($1.2\text{ }\mu\text{g/kg}$), Vital Signs, Safety Assessment
Interim Visits	Every 4 Weeks	Subcutaneous injection of C.E.R.A., Hemoglobin monitoring, Dose titration evaluation, Safety Labs
End of Study	Week 44	Final Hemoglobin assessment, Vital Signs, Exit Interview, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Mean Change in Hb Concentration Between Baseline and the Efficacy Evaluation Period. **Measurement**

Method: Mean change in Hb concentration was calculated as the difference between the time adjusted average of Hb during the efficacy evaluation period (EEP [Week 29 to Week 36]), and the Hb at Baseline (Week 0). A positive change from baseline indicates improvement.

7.2 Secondary & Exploratory Endpoints * Mean Time to Achievement of Response During the EEP. * The Percentage of Participants Whose Hb Concentrations Remained Within the Target Range of \$10.0\$ - \$12.0\$ g/dL Throughout the EEP. * Mean Time Spent by Participants in the Target Range of \$10.0\$ - \$12.0\$ g/dL During the EEP. * Incidence, frequency, and severity of Adverse Events (AEs) to assess safety and tolerability. * Number of dose adjustments required to maintain the individual participant's hemoglobin within the target range.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinician assessment and patient reports at each 4-week visit, specifically monitoring for hypertension and bleeding events.

Serious Adverse Events (SAEs): Reported within 24 hours of investigator awareness, typical of Phase 3 interventional trials.

Data Monitoring Committee (DMC): Internal safety review committee established by Hoffmann-La Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy of hemoglobin correction; Safety Set for all participants who received at least one dose of Mircera. **Sample Size Justification:** A target of 75 participants is planned to provide sufficient descriptive power for the single-arm evaluation of the \$1.2\$ mcg/kg dosing schedule. **Handling of Missing Data:** Last Observation Carried Forward (LOCF) for early study withdrawals regarding hemoglobin maintenance tracking.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to ensure accurate recording of dose adjustments, hemoglobin lab results, and AE/SAE reporting. **Audit Trail:** Secure eCRF locking mechanisms and rigorous data clarification processes managed by the sponsor's clinical data team.

11. Study Identifiers & Governance

Primary ID: ML21524 **Registry Number:** NCT00661388 **Posting ID:** 505575565193260733 **Sponsor/Lead Organization:** Hoffmann-La Roche / Roche **Study Title:** A Study of Monthly Subcutaneous Mircera for the Treatment of Chronic Renal Anemia in Predialysis Patients Not Treated With ESA **Official Regulatory Title:** A Single Arm, Open Label Study to Assess the Efficacy, Safety and Tolerability of Once-monthly Administration of Subcutaneous C.E.R.A. for the Treatment of Chronic Renal Anaemia in Pre-dialysis Patients Not Currently Treated With ESA

12. Clinical Framework (Renal Anemia Specific)

Disease Area: Anemia (Chronic Renal Anemia) **Preferred UMLS Name:** Chronic Kidney Disease-associated Anemia **Semantic Type:** Disease or Syndrome **Definition:** A reduction in the number of red blood cells, the amount of hemoglobin, and/or the volume of packed red blood cells. Clinically, anemia represents a reduction in the oxygen-transporting capacity of a designated volume of blood, resulting from an imbalance between blood loss (through hemorrhage or hemolysis) and blood production. Signs and symptoms of anemia may include pallor of the skin and mucous membranes, shortness of breath, palpitations of the heart, soft systolic murmurs, lethargy, and fatigability. **MeSH Code:** D000740 **HPO Code:** HP:0001903 **SNOMED ID:** 271737000 **SNOMED Hierarchy:** 414022008|Disorder of cellular component of blood|Disorder of cellular component of blood (disorder), 64572001|Disease|Disease (disorder) **Diagnosis Threshold:** Hemoglobin levels typically ≤ 10.0 g/dL in predialysis CKD patients. **Medical Methodology:** Erythropoietin Stimulating Agent (ESA) Therapy / Subcutaneous Mircera (methoxy polyethylene glycol-epoetin beta) injections. **Optimization Target:** Hemoglobin maintained steadily between ≥ 10.0 g/dL and ≤ 12.0 g/dL.

13. Study Procedures & Methodology

Management Pathway: Stepwise dose adjustment of C.E.R.A. based on monthly hemoglobin readings. **Education Protocol (HCP):** Site initiation visits covering the C.E.R.A. titration algorithm and SC administration guidelines. **Education Protocol (Patient):** Counseling on anemia symptoms, blood pressure management, and treatment adherence. **Quality Audit Mechanism:** Continuous safety reviews and regular tracking of out-of-range hemoglobin results to ensure timely dose modifications.

14. Observation & Follow-Up Schedule

Total Duration: 44 weeks **Onsite Visit Cadence:** Every 4 weeks (aligned with the monthly SC administration schedule). **Remote Monitoring Frequency:** Between scheduled dosing visits if clinically warranted for safety. **Checklist Review Interval:** Every 4 weeks prior to each planned dose administration.

15. Sign-off & Approval

Role	Name	Signature	Date	_____	_____	_____	_____
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]	_____	_____	_____	_____
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]	_____	_____	_____	_____
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]	_____	_____	_____	_____